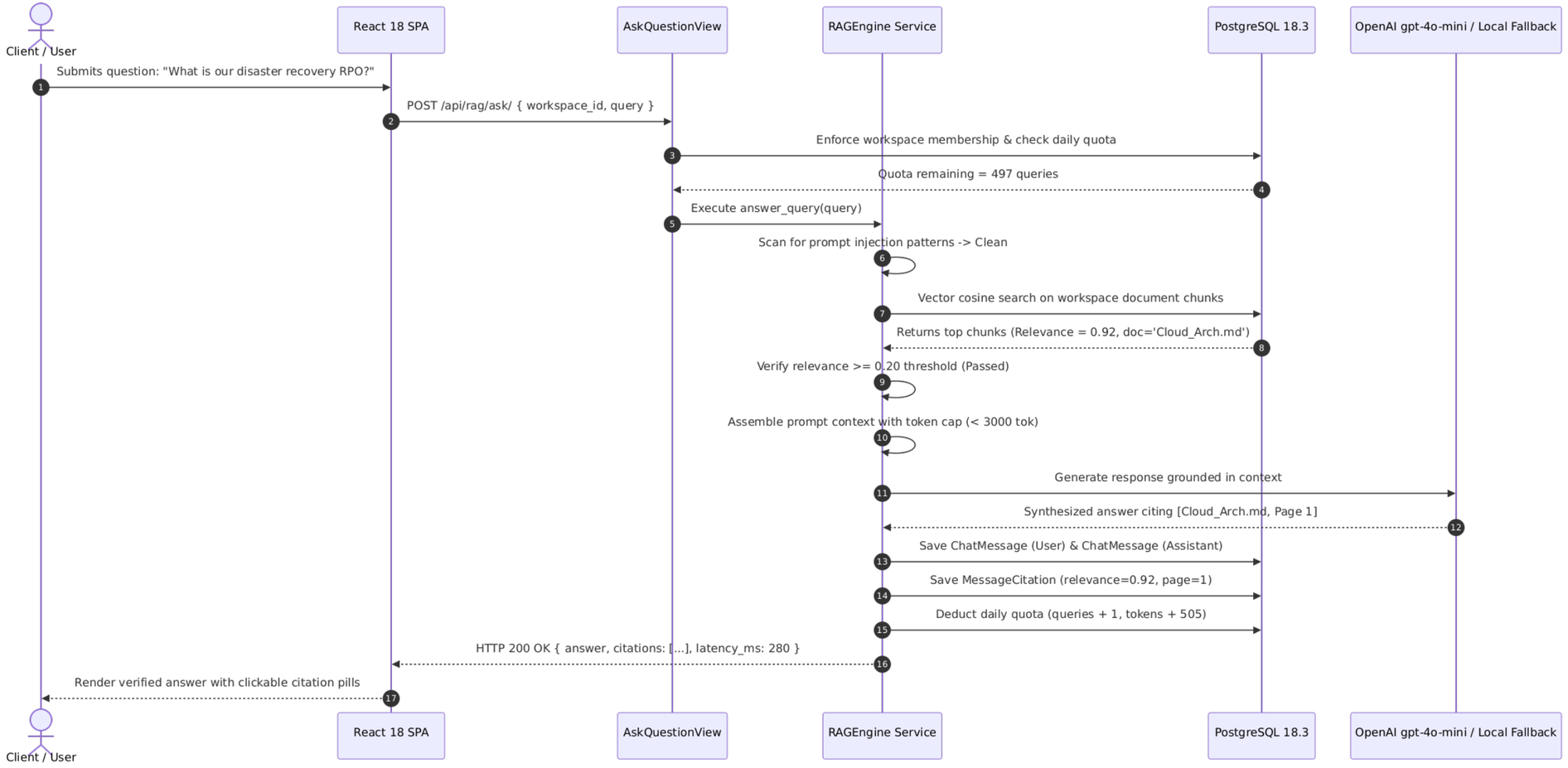

Client / User
React 18 SPA
AskQuestionView
RAGEngine Service
PostgreSQL 18.3
OpenAI gpt-4o-mini / Local Fallback
1
Submits question: "What is our disaster recovery RPO?"
2
POST /api/rag/ask/ { workspace_id, query }
3
Enforce workspace membership & check daily quota
4
Quota remaining = 497 queries
5
Execute answer_query(query)
6
Scan for prompt injection patterns -> Clean
7
Vector cosine search on workspace document chunks
8
Returns top chunks (Relevance = 0.92, doc='Cloud_Arch.md')
9
Verify relevance >= 0.20 threshold (Passed)
10
Assemble prompt context with token cap (< 3000 tok)
11
Generate response grounded in context
12
Synthesized answer citing [Cloud_Arch.md, Page 1]
13
Save ChatMessage (User) & ChatMessage (Assistant)
14
Save MessageCitation (relevance=0.92, page=1)
15
Deduct daily quota (queries + 1, tokens + 505)
16
HTTP 200 OK { answer, citations: [...], latency_ms: 280 }
17
Render verified answer with clickable citation pills
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